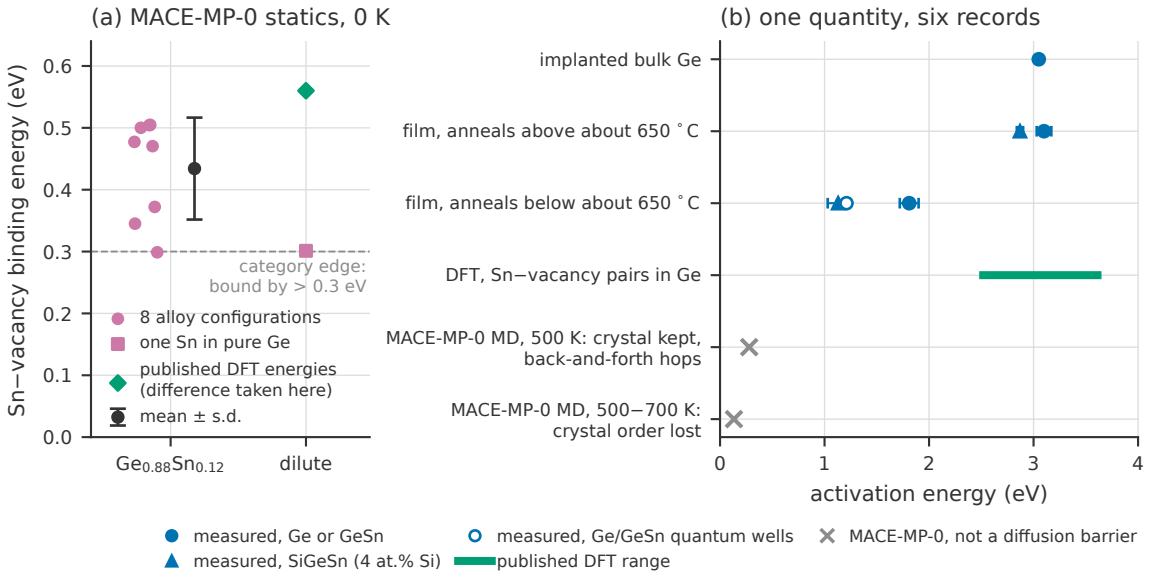
(a) MACE-MP-0 statics, 0 K
Sn−vacancy binding energy (eV)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
category edge:
bound by > 0.3 eV
8 alloy configurations
one Sn in pure Ge
published DFT energies
(difference taken here)
mean ± s.d.
Ge0.88Sn0.12
dilute
(b) one quantity, six records
implanted bulk Ge
film, anneals above about 650 ˚C
film, anneals below about 650 ˚C
DFT, Sn−vacancy pairs in Ge
MACE-MP-0 MD, 500 K: crystal kept,
back-and-forth hops
MACE-MP-0 MD, 500−700 K:
crystal order lost
0
1
2
3
4
activation energy (eV)
measured, Ge or GeSn
measured, Ge/GeSn quantum wells
MACE-MP-0, not a diffusion barrier
measured, SiGeSn (4 at.% Si)
published DFT range